

CS699 Research Progress Report – Team Situational Awareness

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4/6/2025

Over the semester, I have created a data collection plan to apply the Team Situational Awareness research to Agile software development. The data consists of 2 Agile teams with eight individuals each that work for 2-week sprints for four sprints. One team exhibits effective communication patterns and tends to meet their sprint goals, while the other team lacks communication and does not meet all their sprint goals. This data will be used to feed the Situational Awareness models. The data follows a format similar to that of the previous firefighting simulation data. The data fields are as follows: Timestamp, Sender, Recipient, Message, and Sprint. Data is organized by sprint meeting; the ceremonies are daily standup, sprint planning, sprint review, and sprint retrospective. Lastly, another section captures sprint metrics over the course of the program increment. Below are attached sub-sections of the data. The data was sourced from ChatGPT using a notional approach. The data can be generated with a relatively high fidelity; however, it has taken several iterations to get the desired content. Some anomalies still need to be worked out in the set, but it can quickly generate a reasonable number of data points, which is crucial for the training. Some of the problems I ran into were that the meeting times weren't realistic (A meeting could span from 9 am to 4 pm or be held at different times every day during the week). Also, the data would only be around 15 entries, which required me to prompt for a specific number of records. Also, the communication originally described conversations rather than having actual chats, which needed to be reworked. The current iteration has all the required fields with each ceremony having 300 lines for each occurrence, meaning the total for each meeting per sprint is 1200 lines, and the total data for the set is around 4800 records for the positive case and 9600 including both positive and negative. I utilized prompt engineering to improve the dataset from its original state significantly. I will likely find discrepancies in the dataset that will need to be refined as I go along. The current prompts I used for data generation are listed below with the data snippets.

Since the two current datasets are a control group for the positive case and a control group for the negative case, the next steps for the project will require me to derive one additional dataset to enact changes with the SA models. This third dataset will be the test case for SA research that will allow an application of the trained ML model. I must clean and format the data to be compatible with the existing SA models. After I get results, I will need to generate artifacts for content in my thesis from the SA models. Lastly, there will be writing and assembling the thesis document. The outline will consist of around five chapters. Chapter One will explain the previous SA work. Chapters Two and Three will go into the agile process and the data collection

methods for the project. Chapter Four will be on model results and accuracy. Lastly, Chapter Five will go into the future work of how to apply this research to the industry.

Below is the data generated:

Positive case (effective communication, effective goal meeting)

Can you give me an excel spreadsheet with different tabs for each scrum meeting type. Make the data show the communication between team members with timestamps. The communication should be conversational. The team should consist of 8 people with defined roles. The data should show all the meetings over the sprint. The sprint should consist of a 2-week time period. Give data for a total of 4 sprints. The data should be at least 300 lines for each ceremony. Make one more tab that shows the sprint metrics for the team. This team should reach their sprint goals and communicate well.

Daily Standup

	A	B	C	D	E
1	Timestamp	From	To	Message	Sprint
2	2025-03-03 09:02:00	Frank (Tester)	Alice (Scrum Master)	Hey Alice, today I'm working on the login feature and I ran into a small bug.	Sprint 1
3	2025-03-03 09:31:00	Bob (Product Owner)	Charlie (Developer)	Charlie, I finished the API integration yesterday. Moving on to testing today.	Sprint 1
4	2025-03-03 09:32:00	Eve (Developer)	Alice (Scrum Master)	Hey Alice, today I'm working on the login feature and I ran into a small bug.	Sprint 1
5	2025-03-03 09:33:00	Hank (Developer)	Alice (Scrum Master)	Hey Alice, today I'm working on the login feature and I ran into a small bug.	Sprint 1
6	2025-03-03 09:41:00	Dana (Developer)	Alice (Scrum Master)	Hey Alice, today I'm working on the login feature and I ran into a small bug.	Sprint 1
7	2025-03-03 09:42:00	Alice (Scrum Master)	Eve (Developer)	Eve, I finished the API integration yesterday. Moving on to testing today.	Sprint 1
8	2025-03-03 09:44:00	Grace (Developer)	Frank (Tester)	Frank, I finished the API integration yesterday. Moving on to testing today.	Sprint 1
9	2025-03-03 09:51:00	Charlie (Developer)	Grace (Developer)	Morning Grace, still stuck on the deployment issue. Any suggestions?	Sprint 1
10	2025-03-03 10:00:00	Bob (Product Owner)	Eve (Developer)	Hey Eve, today I'm working on the login feature and I ran into a small bug.	Sprint 1
11	2025-03-03 10:10:00	Charlie (Developer)	Hank (Developer)	Morning Hank, still stuck on the deployment issue. Any suggestions?	Sprint 1
12	2025-03-03 10:13:00	Grace (Developer)	Frank (Tester)	Morning Frank, still stuck on the deployment issue. Any suggestions?	Sprint 1
13	2025-03-03 10:15:00	Grace (Developer)	Charlie (Developer)	Morning Charlie, still stuck on the deployment issue. Any suggestions?	Sprint 1
14	2025-03-03 10:16:00	Eve (Developer)	Grace (Developer)	Grace, I finished the API integration yesterday. Moving on to testing today.	Sprint 1
15	2025-03-03 10:24:00	Frank (Tester)	Grace (Developer)	Grace, I finished the API integration yesterday. Moving on to testing today.	Sprint 1
16	2025-03-03 10:36:00	Bob (Product Owner)	Alice (Scrum Master)	Morning Alice, still stuck on the deployment issue. Any suggestions?	Sprint 1
17	2025-03-03 10:45:00	Frank (Tester)	Eve (Developer)	Eve, I might need your help later with some code review.	Sprint 1
18	2025-03-03 10:45:00	Eve (Developer)	Hank (Developer)	Hey Hank, today I'm working on the login feature and I ran into a small bug.	Sprint 1
19	2025-03-03 10:48:00	Dana (Developer)	Eve (Developer)	Eve, I finished the API integration yesterday. Moving on to testing today.	Sprint 1

Sprint Planning

	A	B	C	D	E
1	Timestamp	From	To	Message	Sprint
2	2025-03-11 11:43:00	Frank (Tester)	Grace (Developer)	Grace, do you think we should split this story into smaller tasks?	Sprint 1
3	2025-03-11 10:46:00	Hank (Developer)	Frank (Tester)	Let's clarify the acceptance criteria here, Frank.	Sprint 1
4	2025-03-11 10:19:00	Bob (Product Owner)	Grace (Developer)	Let's clarify the acceptance criteria here, Grace.	Sprint 1
5	2025-03-11 11:52:00	Frank (Tester)	Bob (Product Owner)	Bob, can you take the lead on this component?	Sprint 1
6	2025-03-11 11:13:00	Grace (Developer)	Charlie (Developer)	Let's clarify the acceptance criteria here, Charlie.	Sprint 1
7	2025-03-11 10:42:00	Alice (Scrum Master)	Frank (Tester)	Frank, do you think we should split this story into smaller tasks?	Sprint 1
8	2025-03-11 11:32:00	Charlie (Developer)	Dana (Developer)	I'm not sure about the estimate for this one, Dana. Thoughts?	Sprint 1
9	2025-03-11 10:14:00	Hank (Developer)	Dana (Developer)	I'm not sure about the estimate for this one, Dana. Thoughts?	Sprint 1
10	2025-03-11 10:32:00	Hank (Developer)	Grace (Developer)	I'm not sure about the estimate for this one, Grace. Thoughts?	Sprint 1
11	2025-03-11 11:54:00	Grace (Developer)	Charlie (Developer)	Charlie, can you take the lead on this component?	Sprint 1
12	2025-03-11 11:03:00	Eve (Developer)	Charlie (Developer)	Charlie, can you take the lead on this component?	Sprint 1
13	2025-03-11 10:03:00	Frank (Tester)	Dana (Developer)	Dana, can you take the lead on this component?	Sprint 1
14	2025-03-11 11:42:00	Bob (Product Owner)	Grace (Developer)	I'm not sure about the estimate for this one, Grace. Thoughts?	Sprint 1
15	2025-03-11 10:53:00	Hank (Developer)	Bob (Product Owner)	Bob, do you think we should split this story into smaller tasks?	Sprint 1
16	2025-03-11 11:00:00	Alice (Scrum Master)	Eve (Developer)	Eve, can you take the lead on this component?	Sprint 1

Sprint Planning

	A	B	C	D	E
1	Timestamp	From	To	Message	Sprint
2	2025-03-03 10:10:00	Alice (Scrum Master)	Grace (Developer)	Nice work this sprint, Grace. Let's just polish a few things.	Sprint 1
3	2025-03-03 10:11:00	Bob (Product Owner)	Alice (Scrum Master)	Nice work this sprint, Alice. Let's just polish a few things.	Sprint 1
4	2025-03-03 10:11:00	Hank (Developer)	Dana (Developer)	Dana, do we have feedback from the stakeholders on this story?	Sprint 1
5	2025-03-03 10:11:00	Hank (Developer)	Frank (Tester)	Great job on the UI, Frank. The client loved it.	Sprint 1
6	2025-03-03 10:11:00	Bob (Product Owner)	Dana (Developer)	Great job on the UI, Dana. The client loved it.	Sprint 1
7	2025-03-03 10:13:00	Dana (Developer)	Charlie (Developer)	Great job on the UI, Charlie. The client loved it.	Sprint 1
8	2025-03-03 10:18:00	Dana (Developer)	Bob (Product Owner)	Great job on the UI, Bob. The client loved it.	Sprint 1
9	2025-03-03 10:18:00	Grace (Developer)	Dana (Developer)	Great job on the UI, Dana. The client loved it.	Sprint 1
10	2025-03-03 10:19:00	Bob (Product Owner)	Dana (Developer)	Dana, the feature demo went smoothly. Well done!	Sprint 1
11	2025-03-03 10:19:00	Grace (Developer)	Dana (Developer)	Great job on the UI, Dana. The client loved it.	Sprint 1
12	2025-03-03 10:19:00	Bob (Product Owner)	Dana (Developer)	Dana, do we have feedback from the stakeholders on this story?	Sprint 1
13	2025-03-03 10:21:00	Grace (Developer)	Dana (Developer)	Dana, do we have feedback from the stakeholders on this story?	Sprint 1
14	2025-03-03 10:23:00	Bob (Product Owner)	Dana (Developer)	Dana, do we have feedback from the stakeholders on this story?	Sprint 1
15	2025-03-03 10:29:00	Bob (Product Owner)	Alice (Scrum Master)	Great job on the UI, Alice. The client loved it.	Sprint 1
16	2025-03-03 10:30:00	Frank (Tester)	Eve (Developer)	Nice work this sprint, Eve. Let's just polish a few things.	Sprint 1
17	2025-03-03 10:30:00	Frank (Tester)	Grace (Developer)	Grace, the feature demo went smoothly. Well done!	Sprint 1
18	2025-03-03 10:32:00	Charlie (Developer)	Bob (Product Owner)	Bob, the feature demo went smoothly. Well done!	Sprint 1
19	2025-03-03 10:37:00	Bob (Product Owner)	Hank (Developer)	Hank, do we have feedback from the stakeholders on this story?	Sprint 1

Sprint Retrospective

	A	B	C	D	E
1	Timestamp	From	To	Message	Sprint
2	2025-03-04 11:01:00	Bob (Product Owner)	Frank (Tester)	Next sprint, let's try to improve how we handle blockers, Frank.	Sprint 1
3	2025-03-04 11:02:00	Bob (Product Owner)	Frank (Tester)	Frank, I appreciated your quick testing this sprint.	Sprint 1
4	2025-03-04 11:11:00	Eve (Developer)	Charlie (Developer)	Charlie, I think we collaborated really well this sprint.	Sprint 1
5	2025-03-04 10:26:00	Charlie (Developer)	Grace (Developer)	Grace, I think we collaborated really well this sprint.	Sprint 1
6	2025-03-04 11:01:00	Dana (Developer)	Alice (Scrum Master)	I think daily syncs helped us a lot, right Alice?	Sprint 1
7	2025-03-04 10:52:00	Eve (Developer)	Frank (Tester)	Frank, I think we collaborated really well this sprint.	Sprint 1
8	2025-03-04 10:12:00	Grace (Developer)	Alice (Scrum Master)	Alice, I think we collaborated really well this sprint.	Sprint 1
9	2025-03-04 11:16:00	Eve (Developer)	Charlie (Developer)	Charlie, I think we collaborated really well this sprint.	Sprint 1
10	2025-03-04 10:25:00	Frank (Tester)	Grace (Developer)	Grace, I think we collaborated really well this sprint.	Sprint 1
11	2025-03-04 10:46:00	Hank (Developer)	Eve (Developer)	Eve, I appreciated your quick testing this sprint.	Sprint 1
12	2025-03-04 10:44:00	Grace (Developer)	Bob (Product Owner)	Bob, I appreciated your quick testing this sprint.	Sprint 1
13	2025-03-04 11:33:00	Frank (Tester)	Bob (Product Owner)	Bob, I think we collaborated really well this sprint.	Sprint 1
14	2025-03-04 11:16:00	Frank (Tester)	Alice (Scrum Master)	Next sprint, let's try to improve how we handle blockers, Alice.	Sprint 1
15	2025-03-04 10:43:00	Hank (Developer)	Grace (Developer)	Next sprint, let's try to improve how we handle blockers, Grace.	Sprint 1
16	2025-03-04 11:23:00	Bob (Product Owner)	Frank (Tester)	Next sprint, let's try to improve how we handle blockers, Frank.	Sprint 1
17	2025-03-04 11:10:00	Alice (Scrum Master)	Dana (Developer)	I think daily syncs helped us a lot, right Dana?	Sprint 1
18	2025-03-04 11:07:00	Charlie (Developer)	Bob (Product Owner)	Bob, I appreciated your quick testing this sprint.	Sprint 1
19	2025-03-04 10:32:00	Dana (Developer)	Frank (Tester)	Frank, I appreciated your quick testing this sprint.	Sprint 1
20	2025-03-04 11:12:00	Bob (Product Owner)	Alice (Scrum Master)	I think daily syncs helped us a lot, right Alice?	Sprint 1
21	2025-03-04 10:38:00	Hank (Developer)	Grace (Developer)	I think daily syncs helped us a lot, right Grace?	Sprint 1
22	2025-03-04 11:16:00	Charlie (Developer)	Dana (Developer)	Dana, I appreciated your quick testing this sprint.	Sprint 1
23	2025-03-04 11:08:00	Grace (Developer)	Hank (Developer)	Hank, I appreciated your quick testing this sprint.	Sprint 1
24	2025-03-04 11:20:00	Grace (Developer)	Eve (Developer)	Eve, I think we collaborated really well this sprint.	Sprint 1

Sprint Metrics

	A	B	C	D	E	F	G	H
1	Sprint	Total Story Points Committed	Total Story Points Completed	Team Velocity	Bugs Reported	Bugs Fixed	Team Satisfaction (1-10)	Sprint Goal Achieved
2	Sprint 1	80	80	80	5	2	8	Yes
3	Sprint 2	80	80	80	2	3	10	Yes
4	Sprint 3	80	80	80	4	4	9	Yes
5	Sprint 4	80	80	80	2	5	8	Yes

Negative case (Sub-Par communication, did not meet sprint goals)

Can you give me an excel spreadsheet with different tabs for each scrum meeting type. Make the data show the communication between team members with timestamps. The communication should be conversational. The team should consist of 8 people with defined roles. The data should show all the meetings over the sprint. The sprint should consist of a 2 week time period. Give data for a total of 4 sprints. The data should be at least 300 lines for each ceremony. Make one more tab that shows the sprint metrics for the team. This team should struggle to reach their sprint goals and have some challenges communicating.

Daily Standup

	A	B	C	D	E
1	Timestamp	From	To	Message	Sprint
2	2025-03-03 09:47:00	Alice (Scrum Master)	Dana (Developer)	Hey Dana, I'm blocked but not sure who to ask.	Sprint 1
3	2025-03-03 09:45:00	Bob (Product Owner)	Dana (Developer)	Sorry Dana, I missed the update yesterday. Here's what I did.	Sprint 1
4	2025-03-03 09:47:00	Charlie (Developer)	Bob (Product Owner)	Sorry Bob, I missed the update yesterday. Here's what I did.	Sprint 1
5	2025-03-03 09:45:00	Dana (Developer)	Hank (Developer)	Hey Hank, I'm blocked but not sure who to ask.	Sprint 1
6	2025-03-03 09:50:00	Eve (Developer)	Hank (Developer)	Sorry Hank, I missed the update yesterday. Here's what I did.	Sprint 1
7	2025-03-03 09:59:00	Frank (Tester)	Dana (Developer)	Sorry Dana, I missed the update yesterday. Here's what I did.	Sprint 1
8	2025-03-03 09:35:00	Grace (Developer)	Eve (Developer)	Hey Eve, I'm blocked but not sure who to ask.	Sprint 1
9	2025-03-03 09:50:00	Hank (Developer)	Frank (Tester)	Hey Frank, I'm blocked but not sure who to ask.	Sprint 1
10	2025-03-04 09:20:00	Alice (Scrum Master)	Frank (Tester)	Frank, the deployment failed again... I'll try to look into it.	Sprint 1
11	2025-03-04 09:02:00	Bob (Product Owner)	Alice (Scrum Master)	Hey Alice, I'm blocked but not sure who to ask.	Sprint 1
12	2025-03-04 09:24:00	Charlie (Developer)	Frank (Tester)	Sorry Frank, I missed the update yesterday. Here's what I did.	Sprint 1
13	2025-03-04 09:28:00	Eve (Developer)	Hank (Developer)	Sorry Hank, I missed the update yesterday. Here's what I did.	Sprint 1
14	2025-03-04 09:14:00	Frank (Tester)	Alice (Scrum Master)	Alice, I couldn't get much done yesterday. Still on the same task.	Sprint 1
15	2025-03-04 09:40:00	Grace (Developer)	Charlie (Developer)	Charlie, I couldn't get much done yesterday. Still on the same task.	Sprint 1

Sprint Planning

	A	B	C	D	E
1	Timestamp	From	To	Message	Sprint
2	2025-03-11 11:45:00	Bob (Product Owner)	Eve (Developer)	Eve, are we sure this story is ready? It looks unclear.	Sprint 1
3	2025-03-11 11:49:00	Eve (Developer)	Hank (Developer)	Hank, I think we're overloading the sprint again.	Sprint 1
4	2025-03-11 11:31:00	Eve (Developer)	Grace (Developer)	I'm not confident with the estimates, Grace.	Sprint 1
5	2025-03-11 10:13:00	Dana (Developer)	Eve (Developer)	I'm not confident with the estimates, Eve.	Sprint 1
6	2025-03-11 10:55:00	Eve (Developer)	Frank (Tester)	Frank, I think we're overloading the sprint again.	Sprint 1
7	2025-03-11 11:05:00	Hank (Developer)	Alice (Scrum Master)	I'm not confident with the estimates, Alice.	Sprint 1
8	2025-03-11 10:08:00	Hank (Developer)	Alice (Scrum Master)	Alice, are we sure this story is ready? It looks unclear.	Sprint 1
9	2025-03-11 10:50:00	Bob (Product Owner)	Dana (Developer)	I'm not confident with the estimates, Dana.	Sprint 1
10	2025-03-11 11:36:00	Bob (Product Owner)	Frank (Tester)	Do we have enough clarity on this task, Frank?	Sprint 1
11	2025-03-11 10:10:00	Alice (Scrum Master)	Charlie (Developer)	Charlie, are we sure this story is ready? It looks unclear.	Sprint 1
12	2025-03-11 10:15:00	Alice (Scrum Master)	Eve (Developer)	Do we have enough clarity on this task, Eve?	Sprint 1
13	2025-03-11 11:48:00	Dana (Developer)	Alice (Scrum Master)	Do we have enough clarity on this task, Alice?	Sprint 1
14	2025-03-11 11:21:00	Frank (Tester)	Grace (Developer)	I'm not confident with the estimates, Grace.	Sprint 1
15	2025-03-11 11:06:00	Alice (Scrum Master)	Eve (Developer)	I'm not confident with the estimates, Eve.	Sprint 1
16	2025-03-11 11:55:00	Dana (Developer)	Grace (Developer)	Do we have enough clarity on this task, Grace?	Sprint 1
17	2025-03-11 10:40:00	Alice (Scrum Master)	Eve (Developer)	I'm not confident with the estimates, Eve.	Sprint 1
18	2025-03-11 11:31:00	Hank (Developer)	Alice (Scrum Master)	Alice, I think we're overloading the sprint again.	Sprint 1

Sprint Review

	A	B	C	D	E
1	Timestamp	From	To	Message	Sprint
2	2025-03-05 10:22:00	Eve (Developer)	Frank (Tester)	Frank, there were bugs in the demoed feature.	Sprint 1
3	2025-03-05 10:18:00	Frank (Tester)	Dana (Developer)	The client had questions we weren't ready for, Dana.	Sprint 1
4	2025-03-05 11:10:00	Eve (Developer)	Hank (Developer)	The client had questions we weren't ready for, Hank.	Sprint 1
5	2025-03-05 10:35:00	Charlie (Developer)	Alice (Scrum Master)	Alice, the feedback was mixed this time.	Sprint 1
6	2025-03-05 10:54:00	Frank (Tester)	Grace (Developer)	Grace, not everything got demoed. We ran out of time.	Sprint 1
7	2025-03-05 11:20:00	Eve (Developer)	Bob (Product Owner)	Bob, the feedback was mixed this time.	Sprint 1
8	2025-03-05 10:23:00	Bob (Product Owner)	Eve (Developer)	Eve, there were bugs in the demoed feature.	Sprint 1
9	2025-03-05 11:07:00	Frank (Tester)	Dana (Developer)	Dana, not everything got demoed. We ran out of time.	Sprint 1
10	2025-03-05 10:13:00	Grace (Developer)	Bob (Product Owner)	Bob, not everything got demoed. We ran out of time.	Sprint 1
11	2025-03-05 11:15:00	Alice (Scrum Master)	Hank (Developer)	The client had questions we weren't ready for, Hank.	Sprint 1
12	2025-03-05 10:13:00	Bob (Product Owner)	Alice (Scrum Master)	Alice, not everything got demoed. We ran out of time.	Sprint 1

Sprint Retrospective

	A	B	C	D	E
1	Timestamp	From	To	Message	Sprint
2	2025-03-10 11:08:00	Charlie (Developer)	Grace (Developer)	Too many blockers this sprint, Grace.	Sprint 1
3	2025-03-10 10:57:00	Eve (Developer)	Alice (Scrum Master)	Let's talk about how to better communicate next sprint, Alice.	Sprint 1
4	2025-03-10 11:25:00	Charlie (Developer)	Dana (Developer)	Dana, we missed a few important details again.	Sprint 1
5	2025-03-10 11:41:00	Dana (Developer)	Hank (Developer)	Hank, we need to improve our code reviews.	Sprint 1
6	2025-03-10 10:52:00	Dana (Developer)	Alice (Scrum Master)	Let's talk about how to better communicate next sprint, Alice.	Sprint 1
7	2025-03-10 11:16:00	Dana (Developer)	Bob (Product Owner)	Bob, we need to improve our code reviews.	Sprint 1
8	2025-03-10 11:47:00	Grace (Developer)	Dana (Developer)	Too many blockers this sprint, Dana.	Sprint 1
9	2025-03-10 10:05:00	Frank (Tester)	Charlie (Developer)	Too many blockers this sprint, Charlie.	Sprint 1
10	2025-03-10 11:48:00	Charlie (Developer)	Dana (Developer)	Too many blockers this sprint, Dana.	Sprint 1
11	2025-03-10 10:03:00	Alice (Scrum Master)	Charlie (Developer)	Let's talk about how to better communicate next sprint, Charlie.	Sprint 1
12	2025-03-10 10:36:00	Frank (Tester)	Charlie (Developer)	Let's talk about how to better communicate next sprint, Charlie.	Sprint 1
13	2025-03-10 11:17:00	Alice (Scrum Master)	Eve (Developer)	Eve, we missed a few important details again.	Sprint 1

Sprint Metrics

	A	B	C	D	E	F	G	H
1	Sprint	Total Story Points Committed	Total Story Points Completed	Team Velocity	Bugs Reported	Bugs Fixed	Team Satisfaction (1-10)	Sprint Goal Achieved
2	Sprint 1	80	50	57	8	10	6	No
3	Sprint 2	80	56	65	10	4	4	No
4	Sprint 3	80	57	48	7	8	5	No
5	Sprint 4	80	57	58	10	6	4	No